



# **Playlist Management Using Doubly Linked List**

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# Problem Statement



A music streaming application maintains a playlist where songs can be:

- Added dynamically
- Removed easily
- Reordered anytime
- Played sequentially

The application requires a data structure that provides flexibility and efficient updates.



# Recommended Data Structure



## Doubly Linked List

- A Doubly Linked List stores each song as a node containing:
- Song information
- Pointer to previous song
- Pointer to next song

## Why Doubly Linked List?

- Dynamic memory allocation
- Easy insertion and deletion
- Supports forward and backward navigation



Engineer to Excel

# STRUCTURE DIAGRAM



## Structure Diagram – Playlist using Doubly Linked List

Each song is a node. It has a link to the previous song and the next song, enabling easy addition, deletion, and reordering.



### Add Song

Insert a new song anywhere in the playlist.



### Remove Song

Delete a song easily without affecting other songs.



### Reorder Song

Move songs up or down to change the order.



### Navigate

Move to next or previous song efficiently.

# Operations

| Operation         | Time Complexity |
|-------------------|-----------------|
| Add Song          | $O(1)^*$        |
| Remove Song       | $O(1)^*$        |
| Move/Reorder Song | $O(1)^*$        |
| Search Song       | $O(n)$          |
| Traverse Playlist | $O(n)$          |

\*When the node position is already known.



# Justification



## Why Doubly Linked List?

### Flexibility

- Songs can be inserted anywhere.
- Songs can be deleted easily.
- Playlist size grows dynamically.

### Efficiency

- No shifting of elements.
- Supports Previous and Next buttons.
- Fast updates compared to arrays.



# ADVANTAGES



- Dynamic playlist size
- Efficient insertion and deletion
- Easy song reordering
- Forward and backward navigation
- Better user experience
- Memory allocated as needed



# Conclusion



The **Doubly Linked List** is the most suitable data structure for playlist management because it provides:

- Efficient insertion and deletion
- Flexible song reordering
- Bidirectional navigation
- Dynamic memory usage
- Improved performance for music streaming applications

**THANK YOU**